INCH-POUND

A-A-52208
August 27, 1996
SUPERSEDING
MIL-C-3958C
22 November 1972

COMMERCIAL ITEM DESCRIPTION

CUTTER, CABLE, HAND OPERATED, WITH GROPING HOOK GUIDE

The General Services Administration has authorized the use of this Commercial Item Description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers one type and size hand operated cable cutter with a groping hook guide, having the capacity to cut a 3/8 inch diameter solid steel wire.

1.2 Intended use. The cable cutters covered by this commercial item description are intended for cutting electrically charged barbed wire entanglements and electrical transmission lines not exceeding 5000 volts potential.

2. SALIENT CHARACTERISTICS

2.1 Materials and components. Materials and components shall be as specified herein. Where materials are not definitely specified, they shall be of the quality normally used for this purpose in commercial practice provided the completed cable cutters comply with all provisions of this commercial item description.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other pertinent data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development and Engineering Center, ATTN: SSCNC-WEF, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

2.2 Design and construction. The cable cutter shall be the manufacturer's standard commercially available product similar, but not necessarily restricted to the design of the cutter shown in figures 1 and 2. The cable cutter shall consist of a cutting head, handles, and intermediate links between the handles and jaws. The overall length of the cutter shall be 20 1/2 inches, plus or minus 1/2 inch, and the weight shall be no more than 5 pounds, 2 ounces.

2.2.1 Cutting head. The cutting head shall consist of a pair of jaws, two jaw straps, two strap bolts, and a groping hook guide.

2.2.2 Jaws. The jaws shall be fabricated from a commercial grade fine grain tool steel in accordance with the manufacturer's standard practice. The thickness of each jaw shall be not less than 0.300 inches. The difference in thickness of each jaw at any two points shall not exceed 0.004 inches. The difference in the average thickness of the two mating jaws shall not exceed 0.005 inches. Pivotal means shall be provided on the jaws for equalizing jaw movement and preventing lateral motion of the jaws. The two jaws shall be mounted with no visible sideplay due to excessive clearance between jaws and side bearing surfaces or misalignment of jaw edges. The jaws shall be removable to allow for sharpening of the cutting edges or replacement. Cutting edges of the jaw shall be center cut, similar to figure 1. The bevel hardness of the finished jaws, 0.125 inches from the cutting edges, shall have a Rockwell hardness on the Rockwell C scale of not less than 53 HRC nor greater than 61 HRC, and shall be verified by the test specified in 4.2.2.

2.2.3 Jaw straps. The jaw straps shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. Finished straps shall have a Rockwell hardness on the Rockwell C scale of not less than 43 HRC nor greater than 47 HRC when tested as specified in 4.2.2. The straps shall be either the integral bolt type or the threaded type. When of the integral bolt type, the jaw bolts shall be pressed into one strap and secured to prevent rotation of the bolts and each of the bolts shall be provided with locknuts for securing the jaws. Lock plates, lock washers or other positive means shall be provided to prevent the bolts from becoming loose or rotating. The protruding portion of the bolts shall be held to a minimum to prevent interference in confined areas (see 2.2.5).

2.2.4 Handles. The cutter handles shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. Finished handles shall have a Rockwell hardness on the Rockwell C scale of not less than 43 HRC nor greater than 47 HRC when tested as specified in 4.2.2. When steel tubing is used, the tubing shall be firmly attached to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

preclude loosening during hard usage. The handles shall have a covering of insulation for the protection of the operator against high voltages, and bumpers or stops to prevent the handles from being closed beyond the effective cutting point and pinching the hand of the operator. The handles shall be capable of withstanding a force of 175 pounds on each handle with no permanent deformation in excess of 3/32 inches between the handles, and shall be verified by the test specified in 4.2.1. Handles of fiberglass tubing are acceptable provided they are securely attached, the ends securely plugged, and pass all applicable tests. If fiberglass handles are provided, the dielectric tests in paragraphs 4.2.5 through 4.2.10 are not required.

2.2.5 Bolts. The bolts of the cutter shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. Before assembly, these bolts shall be hardened to a Rockwell hardness on the Rockwell C scale of not less than 30 HRC nor greater than 40 HRC, and shall be verified by the test specified in 4.2.2. The threaded portion of the bolt shall not exceed more than one full thread into the bolt bearing surface of the cutter part when threaded to receive the bolt. The bolts shall have locking means to prevent rotation or loosening of the bolt. When bolt and nut are used, the nut shall be of the prevailing torque locknut type.

2.2.6 Groping hook guide. The groping hook guide shall be no less than 3/32 inches thick, shall consist of a double hook similar to that shown in figures 1 and 2, and shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. The hook guide shall be attached to the cutting head so that the double hook shall be capable of groping for and guiding a wire or cable between the cutter jaws to cut. The double hook shall extend 1 7/8 inches, plus or minus 1/8 inch beyond the cutting head. The hook guide shall withstand a free drop of 3 feet on a hard surface, with no binding of the jaws or no visible evidence of deformation or fractures when tested as specified in 4.2.4.

2.2.7 Intermediate links. The cutter shall have intermediate links that in combination with the handles and jaws, shall provide the cutter with compound movement and mechanical advantage for cutting wire. The intermediate links shall be fabricated from a commercial grade carbon steel in accordance with the manufacturer's standard practice. The intermediate link shall be a separate link or integral with the handle and shall be provided with means for adjustment of the jaws.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

2.2.8 Bumpers. The bumpers on solid handle cutters shall be rubber (see figure 1). The outer surfaces of the rubber bumpers shall be protected with steel washers or rivet heads equal to a minimum of 80 percent of the diameter of the rubber when not compressed. The rubber bumpers shall have a durometer hardness of Shore A 50 to 75 when tested as specified in 4.2.2. The rubber bumpers shall have the capability of withstanding compression to 50 percent of the initial length of the bumper for 48 hours at 80°F ± 10°F with no cracking, splitting, or permanent set of the rubber in excess of 5 percent when tested as specified in 4.2.3. Bumpers for tubular handles shall be solid and integral with the insulation material (see figure 2).

2.2.9 Jaw adjustment. The jaw adjustment shall consist of a 1/4-20 minimum size hardened bolt and locknut, optional with the manufacturer. The jaw adjustment shall be capable of adjusting the spacing between the holes of the tail end of the jaws, a minimum of 1/4 inch.

2.2.10 Insulation. Insulation shall be applied on the handle to obtain thorough adhesion, a smooth surface, and no distortion. Insulation on cutter handles, except tubular handles, shall cover the handle with a minimum of 8 inches of insulation and up to 1 inch of the bumpers plus or minus 1/8 inch. Insulation on tubular handles shall cover the handle up to within 1/2 inch from the jaw end of the tubing and shall have a hand guard projection located 8 inches plus or minus 1/4 inch from the bottom end of the handle to the guard edge nearest the jaw end. Alternatively, in lieu of a hand guard, a hand grip made of the same insulation material may be used, providing it is not less than 4 1/4 inches in length. The insulation shall have a smooth and hard surface with no distortion pinholes, blisters, cuts, waviness or similar imperfections. The handle insulation shall be black and shall have:

(1) A dielectric strength to withstand a 14,000 volt RMS, 60 hertz AC potential for 1 minute, with no leakage or passage of current in excess of 500 microamperes total for both cutter handles when tested as specified in 4.2.5. The insulation shall show no abnormal distortion or blistering after testing in accordance with 4.2.5.

(2) Resistance to moisture absorption so that after immersion in water, the insulation shall withstand a 7,000 volt RMS, 60 hertz AC potential, with no flash over (surface discharge), spark over (air discharge), or breakdown (puncture discharge), when tested as specified in 4.2.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

(3) Resistance to impact so that the insulation of the handles shall withstand the blow of a sharp edged object and also comply with the distortion, blistering, and dielectric requirements in (1) above, when tested as specified in 4.2.7.

(4) Resistance to heat, acid, and alkali with no deterioration of the handle insulation such as blistering, swelling, peeling, becoming spongy or being dissolved, when tested as specified in 4.2.8 or 4.2.9 as applicable.

(5) Resistance to flaming, as self-extinguishing or non-burning of the insulation material when tested as specified in 4.2.10.

2.3 Performance. When tested as specified in 4.2.11, the cutter shall make clean cuts on solid wire under a load of not more than 150 pounds at all points between the innermost and outermost points of the cutting edges without: (1) Binding of movable parts; (2) Cutter parts becoming loose; (3) Indenting or chipping of the jaw edges; (4) Permanent deflection or deformation of the jaw and handle; (5) Evidence of fracture or breakage; (6) Deviation of the initial spacing between jaw cutting edges in excess of 0.010 inches.

2.4 Finish application. Prior to the application of finish, cutter surfaces shall be treated or cleaned to provide a smooth and adherent surface. The exposed metal surfaces of the handle and intermediate links, as well as the cutting head, including bolts, screw nuts, washers, and lock plates shall have a black corrosion resistant finish in accordance with the manufacturer's commercial practice, except in no case shall cadmium plating be used. Cutting edges shall have no finish.

2.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable The contracting officer has the option of accepting or rejecting the product.

2.6 Marking. The cutters shall be permanently and legibly marked with the manufacturer's name or readily identifiable trademark and the following notice:

"WARNING - 5000 VOLTS, MAX.".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

2.7 Workmanship. The completely fabricated cutter shall be free from burrs, slivers, and sharp edges. Cutting edges shall be parallel and in alignment. Parts shall be as specified, free from dents, fractures, breakage or deformation. No parts shall be damaged or impaired in any way and metal parts shall show no evidence of rust or corrosion. There shall be no defects which impair the durability, appearance, or serviceability of the item.

3. REGULATORY REQUIREMENTS

3.1 Contractor recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product that has been sold in the commercial marketplace for over one year, or is the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least two years.

b. The company must have sold 500 items meeting this CID, in the commercial or government marketplace over the past two years.

4.2 Tests.

4.2.1 Deformation. Measure the distance between opposite points, marked and located at the extreme grip end of each handle with the bumpers barely making contact. Restrain the jaws of the cable cutter to prevent the bumpers from making contact and apply a 175 pound load on each handle and release. Check again with the bumpers barely making contact the distance between the two opposite points previously measured to determine compliance with the deformation requirements of 2.2.4.

4.2.2 Hardness. Test the hardness of the jaws, straps, handles, and bolts in accordance with ASTM E 18 to determine compliance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

with the hardness requirements of 2.2.2 through 2.2.5, as applicable. Base the hardness on the average of three readings. Test the hardness of the jaws on the bevel, .125 inches from the cutting edges. Test the hardness of the straps, handles and bolts at any location on the surface. Test the hardness of the rubber bumper in accordance with ASTM D 2240 to determine compliance with the hardness requirements of 2.2.8.

4.2.3 Compressibility. Test the compressibility of the rubber bumper in accordance with ASTM D 395 to determine compliance with the compression requirements of 2.2.8.

4.2.4 Drop. Drop the cutter from a height of 3 feet, plus or minus 1/16 inch, on a concrete floor with the hooks of the groping hook guide facing downward. Repeat the test three times to determine compliance with the hook guide drop requirements of 2.2.6.

4.2.5 Dielectric strength. Immerse the insulated handle of the cutter in 1 percent salt water solution, 2 to 2 1/2 inches from the jaw end edge of the hand guard projection of the handle insulation. If no hand guard protection is furnished, the handles shall be immersed not less than 4 inches on the handle grip. The metal part of the handle shall be one electrode and the salt solution shall be the other electrode, preferably as the ground terminal. The potential between the electrodes shall be raised at a rate of 3,000 volts per second until the final test voltage of 14,000 RMS volts, 60 hertz, AC potential is reached. Maintain the test voltage for a period of 1 minute. During this period, voltage and current readings shall be observed to determine conformance to requirements of 2.2.10 (1). The test equipment shall have a capacity of not less than 1 kilovolt ampere.

4.2.6 Resistance to moisture absorption. Immerse the insulated handle in water for a period of 30 minutes, then wipe dry with a cloth and immediately test as follows. Place two metal foil electrodes on the insulated portion of the handle so their inside edges are 1 inch apart. A potential of 7,000 RMS volts peak, 60 hertz, AC potential shall be applied to the electrodes at a rate of 3,000 volts per second. Apply the test voltage for a period of 1 minute to determine compliance with 2.2.10 (2).

4.2.7 Resistance to impact. Rigidly mount the insulated handle. Swing a hardened steel cutting edge, suspended as a pendulum, to impart a blow to the handle insulation, equivalent to 5 foot pounds of energy. The cutting edge shall strike the insulation at right angles to the longitudinal axis of the handle to determine compliance with 2.2.10 (3). The cutting edge shall consist of a 90 degree wedge, 0.010 inches wide and 1 inch long.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

4.2.8 Resistance to acid. Immerse and soak the insulated handle in a 20 percent sulfuric acid solution for a period of 1 hour at 150°F minimum; then remove from the acid solution, rinse thoroughly in distilled water and wipe dry. The insulation shall be examined to determine compliance with the applicable requirements of 2.2.10 (4).

4.2.9 Resistance to alkali. Immerse and soak the insulated handle in a 40 percent sodium hydroxide solution for a period of 1 hour at 150°F minimum; then remove from the alkali solution, rinse thoroughly in distilled water and wipe dry. The insulation shall be examined to determine compliance with the applicable requirements of 2.2.10 (4).

4.2.10 Resistance to flaming. Test the insulation of the handle for flammability resistance in accordance with ASTM D 635 to determine compliance with the applicable requirements of 2.2.10 (5).

4.2.11 Performance. Make 6 complete cuts with the cable cutter on a commercially available carbon hard-drawn solid wire, 3/8 inches in diameter, under a load of not more than 150 pounds. Apply the load 2 inches plus or minus 1/8 inch from the end of each handle. Make each cut at separate points from the innermost to the outermost points of the cutting edges to determine compliance with the performance requirements of 2.3.

5. PACKAGING

5.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number and date of this document.
b. Packaging requirements (see 5.1).

6.2 National Stock Number. The following national stock number corresponds to this CID: 5110-00-541-6730.

6.3 Sources for non-Government association documents.

ASTM

D 395 - Standard Test Methods for Rubber Property - Compression Set

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

D 635 - Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position
D 2240 - Standard Test Method for Rubber Property - Durometer Hardness
E 18 - Methods of Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 380 - Use of the International System of Units (SI) (The Modernized Metric System)

ASTM documents are available from:

The American Society for Testing and Materials (ASTM)
100 Barr Harbor Drive
West Conshohocken, PA 19428

6.4 Sources of supply. The manufacturer whose products are known to meet the CID requirements is listed below. However, competition is not limited to this company:

Ameritool
Bar-Es Division
16741 West Park Circle Drive
Chagrin Falls, OH 44023

MILITARY INTERESTS:

Custodians

Army - GL
Navy - AS
Air Force - 99

Review activities

Navy - CG, MC
Air Force - 03, 84

CIVIL AGENCY COORDINATING ACTIVITY:

DOI - BLM
GSA - FSS

PREPARING ACTIVITY:

Army - GL

Project No. 5110-0351

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-52208	960827

3. DOCUMENT TITLE
Cutter, Cable, Hand Operated, with Groping Hook Guide

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
U.S. Army Natick RD&E Center	508-233-5175	256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SSCNC-WEF Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T09:06Z
Check the source to verify that this is the current version before use.

NATICK OP1 1 MAR 96 NATICK OP1 1 Jul 01 IS OBSOLETE